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Efficient Cleavage of Lignin–Carbohydrate Complexes and Ultrafast Extraction of Lignin Oligomers from Wood Biomass by Microwave-Assisted Treatment with Deep Eutectic Solvent

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Lignocellulosic biomass is an abundant and renewable resource for the production of biobased value-added fuels, chemicals, and materials, but its effective exploitation by an energy-efficient and environmentally friendly strategy remains a challenge. Herein, a facile approach for efficiently cleaving lignin–carbohydrate complexes and ultrafast fractionation of components from wood by microwave-assisted treatment with deep eutectic solvent is reported. The solvent was composed of sustainable choline chloride and oxalic acid dihydrate, and showed a hydrogen-bond acidity of 1.31. Efficient fractionation of lignocellulose with the solvent was realized by heating at 80 °C under 800 W microwave irradiation for 3 min. The extracted lignin showed a low molecular weight of 913, a low polydispersity of 1.25, and consisted of lignin oligomers with high purity (ca. 96 %), and thus shows potential in downstream production of aromatic chemicals. The other dissolved matter mainly comprised glucose, xylose, and hydroxymethylfurfural. The undissolved material was cellulose with crystal I structure and a crystallinity of approximately 75 %, which can be used for fabricating nanocellulose. Therefore, this work promotes an ultrafast lignin-first biorefinery approach while simultaneously keeping the undissolved cellulose available for further utilization. This work is expected to contribute to improving the economics of overall biorefining of lignocellulosic biomass.

Introduction

In light of the gradual depletion of fossil fuels and emerging environmental concerns, there is a strong incentive to exploit alternative energy sources.[1,2] Lignocellulosic biomass, mainly present in the cell walls of plant resources such as trees, is an abundant and renewable resource for the production of bio-derived value-added fuels, chemicals, and materials.[3–7] If the separation, extraction, and subsequent chemical transformation of lignocellulosic biomass into these products are performed rapidly by a simple reaction pathway, the maximum value of the biomass feedstock as a whole can be achieved.[7]

The fractionation of lignocellulosic biomass in a green and efficient way is essential for decreasing negative effects on the environment. Although numerous methods for refining lignocellulose have been developed in recent years, it still remains a serious challenge to break the recalcitrance of biomass through an energetically efficient and environmentally friendly process.[8,9] Biomass recalcitrance generally arises from lignin–carbohydrate complexes (LCCs), in which lignin is mainly covalently bonded to hemicellulose.[10,11] The traditional methods to break the LCC structure involve physical, chemical, physicochemical, and biological pretreatments such as steam explosion, ammonia fiber expansion, acid-catalyzed prehydrolysis, and microorganism digestion.[12–15] Besides, organosolv and ionic liquid pretreatments are recognized as efficient methods for fractionation of lignocellulose components following the concepts of green chemistry and biorefinery.[16–19] In organosolv protocols, the mechanism of breaking LCC linkages involves hemicellulose hydrolysis, and simultaneously lignin is dissolved in the organosolv liquor.[20,21] Ionic liquids open a new window for lignocellulose fractionation because they can partially or entirely dissolve lignin and carbohydrates.[22] Several new and cheap ionic liquids were also found to be effective for biomass pretreatment and conversion.[23–27]

Recently, deep eutectic solvents (DESs), which have the characteristics of both ionic liquids and organic solvents, have emerged as a new generation of green solvents with low cost, high atom efficiency, and techno-economically applicable characteristics. DESs are generally prepared by the self-association of hydrogen-bond donors (HBD) and hydrogen-bond acceptors (HBA), which form a liquid driven by strong hydrogen-bonding

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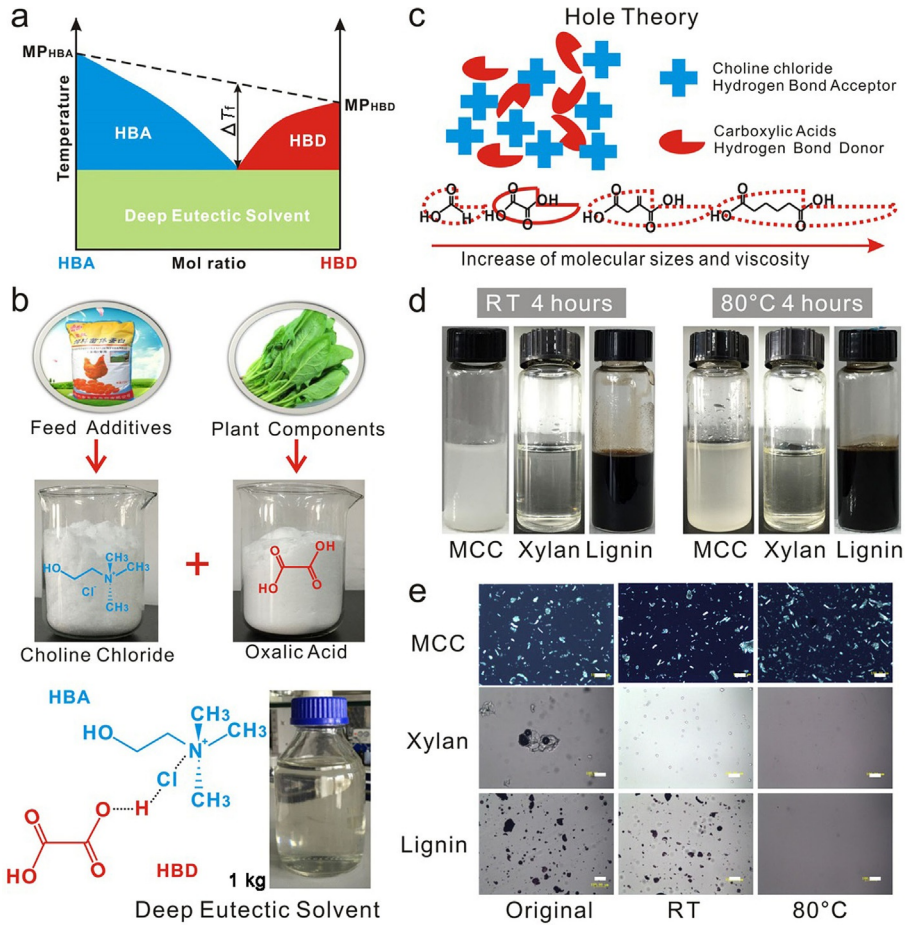


Figure 1. a) Schematic representation of the eutectic point between HBA and HBD. b) The DES composed of choline chloride and oxalic acid dihydrate. c) Model illustration of Hole theory. d) Digital photographs and e) optical micrographs showing the solubility of MCC, xylan, and lignin. The scale bar in e) is 100 μm.

interactions that reduce the ability of the parent compounds to crystallize (Figure 1 a).[28,29] Generally, a DES is obtained by mixing a quaternary ammonium halide salt with a neutral organic HBD to form a complex with halide ions and solvent molecules.[30–32] DESs exhibit several advantages, such as easy and cheap fabrication, no need for purification, and insensitivity to water, and have been widely accepted. With these characteristics, DESs have been widely utilized in many fields,[33–37] such as metal processing[33] and synthesis media or templates.[34,35] DESs also have effects on delignification and can be used in the extraction of bioactives[38,39] and biomass refinery.[40–47] Francisco et al. reported that a DES was promising for wheat straw processing because lignin was partially dissolved.[41] Sirviö et al. showed that acidic DESs can be hydrolytic media for lignocellulose pretreatment and produce cellulose nanocrystals.[36,42,43] Xu et al. used a DES to enhance the accessibility of corn stover for butanol production.[46] Vasco et al. reported that low-molecular-weight lignin could be extracted from wood in DES at 145–180 °C.[47] However, to the best of our knowledge, the influence of applying microwave-assisted DES to improving the fractionation of lignocellulosic biomass and fabricating high-value chemicals has not been studied. Detailed studies to disclose the mechanism of cleaving interlinkages of the LCC structure by DES and conversion to important chemicals remain limited and less well understood.

In this study, aimed at seeking a DES for targeted LCC cleavage, sustainable choline chloride (ChCl) and oxalic acid dihydrate were selected under the guidance of solvatochromic parameters to prepare a DES befitting wood lignocellulose (WL) refinery. Process intensification through the integration of dissolution, fractionation, and conversion of WL to lignin oligomers, cellulose, and saccharides was accomplished in one pot in 3 min. Thus, through this facile pretreatment process, WL components could be successfully fractionated and transformed into valuable products for sustainable chemistry and energy utilization.

Results and Discussion

Preparation of a befitting DES for WL fractionation

There are manifold combinations of HBDs and HBAs to prepare DESs. In preliminary experiments, ChCl was selected to couple with three types of HBDs (amines, carboxylic acids, and alcohols) to obtain three kinds of DESs (Figure S1 in the Supporting Information). All of them showed good solubility of xylan

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and lignin but poor solubility of microcrystalline cellulose (MCC) (Figure S2). Kamlet–Taft solvatochromic parameters were used to reflect the chemical properties of these DESs. The parameters α, β, and π^* represent hydrogen-bond acidity, hydrogen-bond basicity, and dipolarity/polarizability of the solvents, respectively. This revealed that the DES with ChCl and oxalic acid dihydrate has the highest hydrogen-bond acidity ($\alpha =$ 1.31), which is considered to be conducive to breaking the LCC structure of WL (Table S1). The ChCl/oxalic acid dihydrate DES also showed the best performance in the treatment of WL (Figure S3). ChCl and oxalic acid dihydrate are available as animal feed additives and natural products of plants, and both are abundant, renewable, and inexpensive (Figure 1 b). Furthermore, Hole theory (Figure 1 c) suggests the viscosity of DESs is related to the free volume and probability of finding holes for solvent molecules/ions to move into.[29] Oxalic acid dihydrate with a short molecular chain was deemed a befitting HBD for improving the accessibility of WL and inducing disintegration. After mixing ChCl and oxalic acid dihydrate at 80 °C under reduced pressure for 1 h, a clear DES was formed at room temperature. A molar ratio of 1:1 was verified to be a priority in obtaining the optimum DES (Figure S4). No waste was generated in the preparation process, and the atom economy was nearly 100% with an environmental factor of 0. The preparation of this DES was facile, and it could easily be produced on a kg scale.

The ChCl/oxalic acid dihydrate DES contains abundant nonsymmetrical chloride ions and carboxylic acid molecules, and it has the features of both ionic liquids and organic solvents. To survey the effect of the DES on WL components, MCC, xylan, and alkali lignin were selected as the model compounds to test their solubility in DES. The DES showed good solubility of xylan and lignin at room temperature, but poor solubility of MCC (Figure 1 d). This finding was possibly a result of the effect of chloride ions in breaking the intramolecular hydrogen-bond network, and the oxalic acid solvent contributes to dissolving lignin fractions and polysaccharides. At 80 °C in an oil bath, the dissolution behavior of xylan and lignin seemed more pronounced than at room temperature, and the solubility of MCC was slightly improved. Optical micrographs were recorded to observe the changes in the model compounds (Figure 1 e). The images indicate that, at room temperature, xylan and lignin were partially dissolved and MCC remained undissolved. However, after 4 h in an oil bath at 80 °C, xylan and lignin were almost entirely dissolved, and MCC was still insoluble. These results suggest a good potential for using this DES for the selective fractionation of hemicellulose and lignin from WL.

Efficient fractionation of WL by microwave-assisted DES treatment

Given the above survey results, the DES was used to fractionate WL. WL was mixed with ChCl/oxalic acid dihydrate DES in a mass ratio of 1:20. The DES processing modes of oil-bath heating (80 and 110 °C for 9 h; $DES_{OB80\text{-}9}$ and $DES_{OB110\text{-}9}$, respectively) and microwave-assisted heating (80 °C, microwave power 800 W; DES_{Mw80}) were conducted and compared. As shown in Figure 2 b, after heating at 80 °C in an oil bath for 9 h, the solution had a light brown color, suggesting that only a small portion of lignin had been extracted. If the oil bath temperature was increased to 110 °C, the color of the solution turned dark brown, which suggested a high content of extracted lignin. However, these processing conditions were still unsatisfactory from the viewpoint of efficiency and energy consumption. When microwave irradiation was applied during the DES treatment process, it only took 3 min (heating up for 2 min and retention for 1 min; $DES_{Mw80\text{-}3}$) to obtain a similar

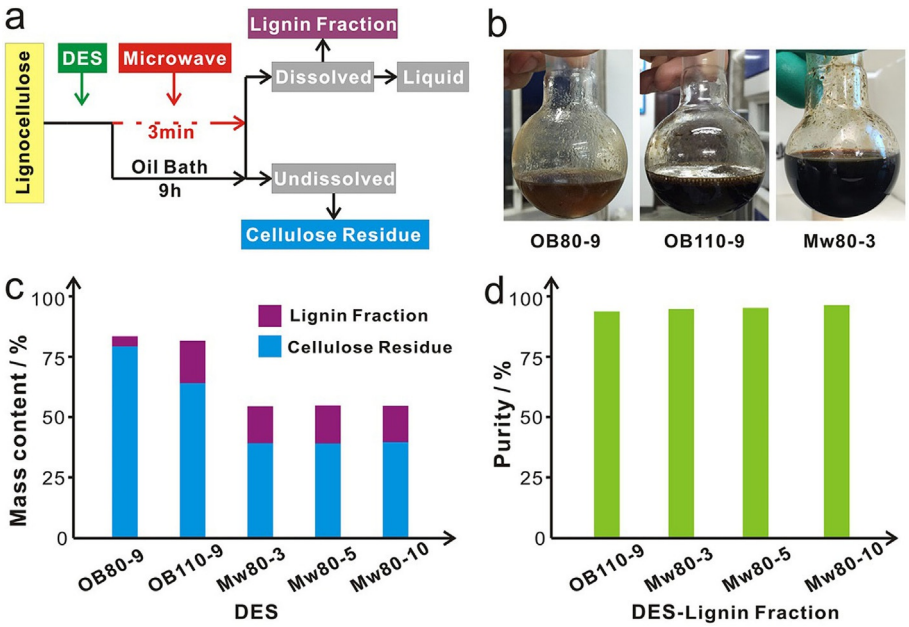


Figure 2. a) Schematic diagram showing the fractionation products of WL by DES. b) Digital photographs of WL treated by DES under different conditions. c) Mass content of LF and CR. d) Content of the acid-insoluble lignin in the DES-extracted lignin.

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dark brown solution to that from oil-bath heating at 110 °C for 9 h. This result shows that microwave radiation can maximize the ionic characteristics and increase the molecular polarity of the DES; therefore, both the temperature and processing time can be decreased. A combination of microwave irradiation and DES can realize efficient fractionation of WL.

After DES treatment, the WL was fractionated into dissolved liquid fractions and undissolved solid residues (Figure 2a). From the above-mentioned results, the dissolved liquid part ought to be the fractions of lignin and hemicellulose, and the undissolved solid part is the cellulosic residue (CR, Figure S5). CR was easily isolated by vacuum filtration, and the lignin fraction (LF) was obtained by a solvent-evaporation method. The mass contents of CR and LF corresponding to each set of processing conditions were calculated (Figure 2c). The results reveal that a higher temperature benefits the separation of lignin, and approximately 17.5% LF (90% of the initial lignin content of 19.3%) was fractionated if the temperature of the oil bath was increased to 110 °C. The content of CR was 64.1%, which was higher than that of either α-cellulose (43.5%) or hemicellulose (35.3%) in WL. The microwave-assisted DES treatment also showed good results in lignin extraction. During microwave-assisted DES treatment for 3 min, 15.4% LF was extracted, which accounts for 80% of the total lignin. The CR content decreased compared with that at 110 °C, possibly because of partial depolymerization of amorphous cellulose under microwave conditions. When the microwave irradiation time was extended to 5 min (heating up for 2 min and retention for 3 min) and 10 min (heating up for 2 min and retention for 8 min), the extracted LFs were 15.8 and 15.1%, and the CRs were 39.2 and 39.6%, respectively. These results indicate that extending the microwave irradiation time has little effect on further fractionation of WL components. The content of acid-insoluble lignin was determined to denote the percentage lignin purity in LF (Figure 2d). LF with a high purity (93.7–96.3%) of acid-insoluble lignin could be obtained by DES fractionation. Both the CR contents were less than the α-cellulose content in WL, and this suggests that small amounts of amorphous α-cellulose were dissolved by the treatments. Only 3.7–4.4% lignin remained in the CR (Figure S6).

Cleavage of LCC linkages and transformation of substructures

A major restriction for fractionation of WL lies in the recalcitrance of LCC (for the LCC structure, see Figures 3a and S7), which is formed by covalent bonding of lignin to hemicellulose containing benzyl ester, benzyl ether, and phenyl glycoside groups and further cross-linking through strong hydrogen-bonding interactions. The strong hydrogen-bonding interactions in WL may be weakened because of competing hydrogen-bond formation between the chloride ions of the DES and hydroxyl groups in the carbohydrates and lignin, which breaks the LCC linkages at the locations indicated by the arrows in Figure 3a. Thus, separation of lignin and hemicellulose with DES can occur by the cleavage of LCC linkages.

For more insight into the nature of DES-extracted LF, detailed structures were identified by ^{1}H–^{13}C 2D HSQC NMR spectroscopy (Table S2). Milled wood lignin (MWL) was prepared as control to compare with $DES_{OB110\text{-}9}$ and $DES_{Mw80\text{-}3}$ LFs. In the side-chain region of MWL (Figure 3b), the cross-signals of methoxyl groups (OMe, $\delta_C/\delta_H = 56.2/3.73$) and β-O-4 substructures (A) were the most prominent. C_α–H_α correlations were observed at $\delta_C/\delta_H = 72.52/4.83$ (A and A′). The C_β–H_β correlations corresponding to the *erythro* and *threo* forms of syringyl (S) β-O-4 substructures were distinguished at $\delta_C/\delta_H = 83.09/4.45$ and 86.60/4.11. The C_γ–H_γ correlations in β-O-4 substructures were observed at $\delta_C/\delta_H = 58.98–60.56/3.40–3.72$. The correlations shifted to $\delta_C/\delta_H = 80.11/4.41$ in β-O-4 substructures linked to guaiacyl (G)/*p*-hydroxyphenyl (H) lignin units. The cross-signals of resinol structures (B) were observed with their C_α–H_α, C_β–H_β, and double C_γ–H_γ correlations. Phenylcoumaran substructures (C) were indicated by the signals of C_α–H_α and C_β–H_β correlations at $\delta_C/\delta_H = 87.48/5.44$ and 53.62/3.47. There was also a small amount of signals corresponding to the *p*-hydroxycinnamyl alcohol end groups (I, Figure S8). The cross-signals of benzyl ether could be detected at $\delta_C/\delta_H = 81.3/4.65$. The cross-signals of xylans in benzyl ester structures were detected through the C_2–H_2, C_3–H_3, C_4–H_4, and C_5–H_5 correlations (β-D-xylopyranoside, labelled as X2, X3, X4, and X5). The cross-signals of 3-*O*-acetyl-β-D-xylopyranoside and 2-*O*-acetyl-β-D-xylopyranoside were detected ($X3_3$ and $X2_2$), and prove the existence of typical lignin xylans in MWL. The cross-signals in the aromatic region ($\delta_C/\delta_H = 100–135/5.5–8.5$) correspond to the aromatic rings of different lignin units. In MWL, the signals of S-type and G-type units were clear (Figure 3c). Signals at $\delta_C/\delta_H = 106.7/7.36$ and 7.21 correspond to $C_{2,6}$–$H_{2,6}$ correlations in C-α-oxidized S-type lignin units. Other distinct signals were assigned to *p*-hydroxybenzoate substructures (PB) and cinnamaldehyde end groups (J).

In the side-chain regions and aromatic regions of $DES_{OB110\text{-}9}$ LF (Figure 3d and e), the cross-signals of X2, X3, X4, C, and J could not be detected anymore, but those of benzyl ether and PB still existed. The results reveal that some primary linkages between xylans and lignin were cleaved by DES treatment. When microwave irradiation was applied during DES treatment at 80 °C for 3 min, both the cross-signals of benzyl ether and xylans disappeared (Figure 3f). Typical structures of A_α, A_β, and C also disappeared. This might correspond to the cleavage of ether bonds in these structures, leaving the B structures with carbon–carbon bonds. This result conformed to those in the recent literature.[47] In the aromatic regions of $DES_{Mw80\text{-}3}$ LF (Figure 3g), PB substructures showed reduced cross-signals, and the J cross-signals completely disappeared. However, the signals of G-type units were observed in the aromatic region of both $DES_{OB110\text{-}9}$ and $DES_{Mw80\text{-}3}$ LF. This revealed that condensation of lignin occurred during the extraction process. These results reveal that the short-time microwave-assisted DES treatment also played an effective role in breaking the LCC linkages and fractionated the lignin and xylans.

The molecular weight of $DES_{Mw80\text{-}3}$ LF was determined by gel-permeation chromatography (GPC, Figures S9 and S10). The weight-average (M_w) and number-average molecular

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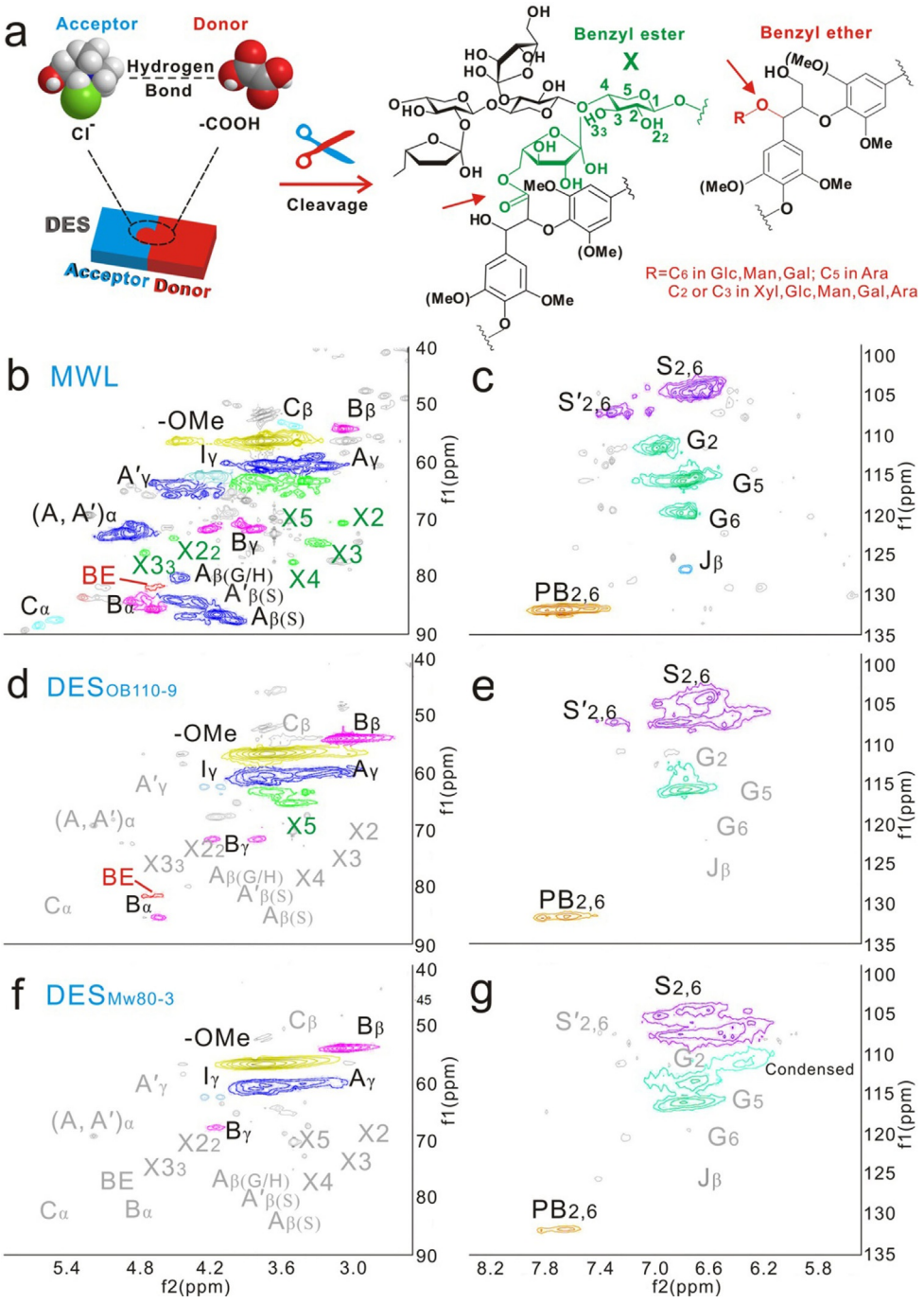


Figure 3. a) Supposed cleavage sites (red arrows) of typical LCC benzyl ester structures formed with xylans and benzyl ether structures formed with C_5 or C_6 saccharides in hemicellulose) by DES. 2D HSQC NMR spectra of the structures in the side-chain regions and aromatic regions of b), c) MWL, d), e) $DES_{OB110-9}$ LF, and f), g) DES_{Mw80-3} LF.

weights (M_n) are listed in Table 1. The polydispersity indices of $DES_{OB110-9}$, DES_{Mw80-3}, DES_{Mw80-5}, and $DES_{Mw80-10}$ LF were calculated as 1.18, 1.25, 1.21, and 1.24, respectively. The results reveal that the M_w of DES-extracted LF showed a smaller and narrower uniform distribution compared with those from other treatments (Table S3).

The ratio of S-type and G-type lignin units in MWL and DES_{Mw80-3} LF was analyzed by pyrolysis (Py)/GC–MS (Figure 4a and b). From the determination of relative abundance, the S/G

Table 1. M_w, M_n, and polydispersity (M_w/M_n) of the LFs from different DES treatments.

	$DES_{OB110-9}$	DES_{Mw80-3}	DES_{Mw80-5}	$DES_{Mw80-10}$
M_w	804	913	847	900
M_n	681	733	698	724
M_w/M_n	1.18	1.25	1.21	1.24

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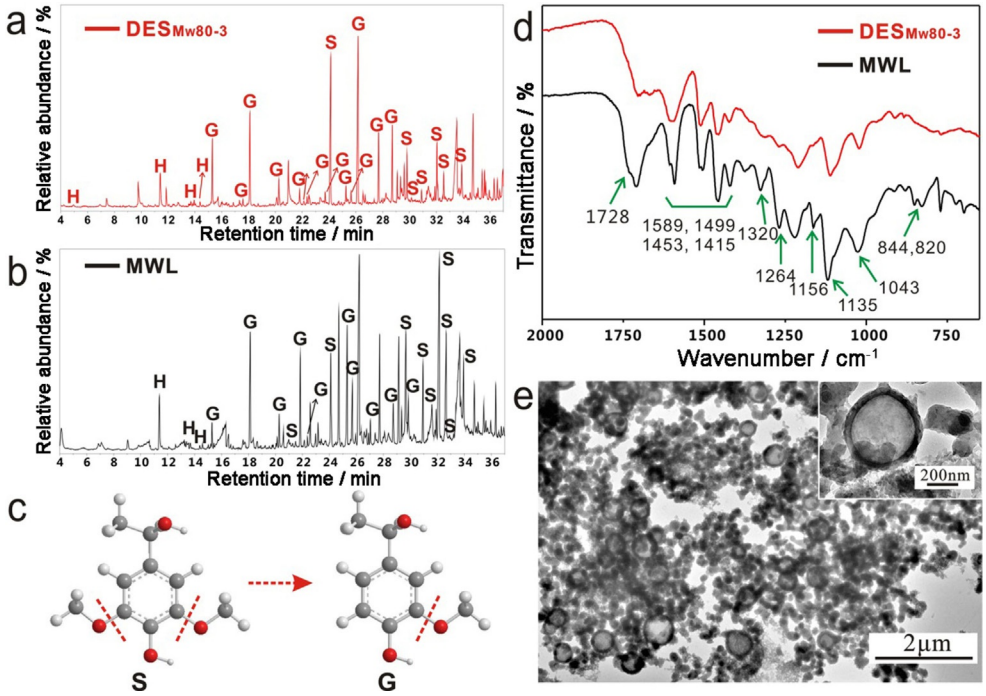


Figure 4. Py/GC–MS spectra of a) DES_{Mw80-3} LF and b) MWL. c) Molecular structure of S-type and G-type lignin units. d) FTIR spectra of DES_{Mw80-3} LF and MWL. e) TEM images of the regenerated DES_{Mw80-3} LF.

ratio was calculated.[48] The S/G ratio of MWL was 1.41, and that of DES_{Mw80-3} LF was 0.69 (Tables S4 and S5). The G-type units in DES_{Mw80-3} showed an obvious increase, which may be because the demethoxy effect of DES results in a transformation from S-type to G-type units (Figure 4c). The transformation process was also verified by FTIR analysis (Figure 4d). The absorption bands at 1589, 1499, 1453, and 1415 cm^{-1} indicate that MWL and DES_{Mw80-3} LF have similar core structures (Table S6). The absorption bands at 1728 and 1156 cm^{-1} were assigned to C=O stretching in unconjugated ketone and carboxyl groups and C–O stretching in ester groups, respectively. The disappearance of these two absorption bands indicates cleavage of benzyl esters. Absorption bands at 1320 and 1135 cm^{-1} were assigned to S-type ring breathing with C–O stretching and aromatic C–H in-plane deformation of S-type units, respectively.[49] The absence of absorption bands at 1320 and 1264 cm^{-1} shows a decrease of S-type units. These results are consistent with those of Py/GC–MS.

CR, saccharides, and hydroxymethylfurfural

As another product of WL after DES treatment, CR was characterized by the crystalline structure, relative crystallinity, degree of polymerization (DP), and thermal stability. The crystalline structure of CR was classified as crystal I type by XRD (Figure S11), and the relative crystallinity approached 75% (Figure 5b). This reveals that after the DES treatment, the CR in WL remained a crystal I-type cellulose with high crystallinity. The thermal stability of CR was similar to that of the original cellulose (Figure 5d). The above results suggest that the DES-derived CR is a promising feedstock to generate cellulosic nanomaterials such as nanocellulose. By conducting a 1000 W ultrasonication treatment of the CR solution (concentration: 0.5 wt%) for 20 min,[50] a homogeneous solution of nanofibrillated cellulose can be easily obtained (Figures 5a and S12). TEM images show that the obtained nanofibrillated cellulose had a similar morphology and uniform diameter distributed within 8–14 nm (Figure S13). The DP of cellulose after DES treatment decreased to 285 from its original value of 1288 (Figure 5c). This reveals that the microwave-assisted DES treatment has a partial depolymerizing effect on cellulose, and such CR may be applied for bioethanol production.

After the removal of LF and CR, the remaining solution (Figure S14) was vacuum-distilled and quantitatively analyzed by high-performance anion-exchange chromatography (HPAEC). This showed mainly two kinds of abundant saccharides, that is, glucose and xylose, and small amounts of rhamnose and galactose (Table 2). These saccharides were deemed to form from hemicellulose and amorphous cellulose in WL through DES hydrolysis. The content of glucose and xylose accounted for 22.3 and 34.8 wt% of their original contents, respectively. Hydroxymethylfurfural (HMF, 1.07 wt% of the WL) was also detected, which suggests that transformation of glucose into HMF might

Table 2. Comparison of the DES-extracted saccharides with those in WL.

	Glucose	Xylose	Rhamnose	Galactose	Arabinose	HMF
WL [%]	22.7[a]	20.0[a]	0.3[a]	0.6[a]	0.3[a]	–
DES-extracted [%]	5.06[a]	6.96[a]	0.093[a]	0.009[a]	n.d.[b]	1.07[a]

[a] Mass percentage relative to WL. [b] Not detected.

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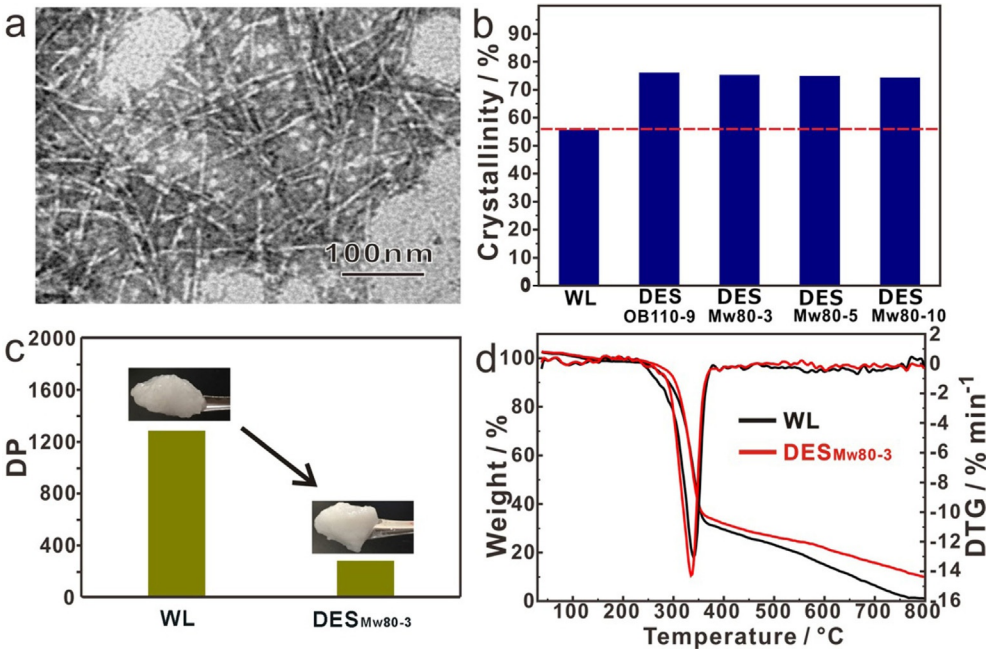


Figure 5. a) Nanocellulose obtained by ultrasonic nanofibrillation of the CR. b) Relative crystallinity, c) DP, and d) thermogravimetric (TG) and derivative thermogravimetric (DTG) curves of WL and DES-treated CR.

have occurred (Figure S15). From the perspective of reuse of the waste liquor containing abundant saccharides, the existing saccharides need not be isolated from the liquor, and they can be directly transformed into carbon materials through a simple hydrothermal treatment (Figure S16).

The DES can be easily recycled by a distillation process. The ^{1}H NMR spectrum of the recycled DES (Figure S17) is similar to that of the original DES, except that the peak of oxalic acid is somewhat weaker, that is, oxalic acid may be consumed during the fractionation process. When the recycled DES was used to once again pretreat the WL, 12.8% LF (accounting for 66.3% of the lignin in WL) and 46.3% CR were extracted, that is, the recycled DES remains useful and effective.

Conclusions

A powerful ChCl/oxalic acid dihydrate deep eutectic solvent (DES) with a hydrogen-bond acidity of 1.31 was prepared and applied as a new green solvent for wood lignocellulose (WL) fractionation. The combination of DES and microwave irradiation had a significant synergetic effect on efficiently cleaving the lignin–carbohydrate complex (LCC) structure and ultrafast fractionation of WL. The extracted lignin fraction (LF) was of low molecular weight (913), low polydispersity (1.25), and high purity. This revealed that the DES-extracted lignin could be used as feedstock for downstream production of aromatic chemicals. Glucose, xylose, and hydroxymethylfurfural (HMF) were generated during the one-pot process. The undissolved cellulose was of crystal I structure with a crystallinity around 75% and a degree of polymerization (DP) of 285, and thus shows potential for producing nanocellulose and bioethanol. The integration of DES and microwave treatment promotes an efficient lignin oligomers-first biorefinery approach while keeping the dissolved saccharides and undissolved cellulose available for further utilization.

Experimental Section

Materials: Poplar wood flour (80 mesh) was dried at 103 °C for 24 h and used as the WL. MWL was prepared as control according to Note S1 in the Supporting Information. ChCl, oxalic acid dihydrate, urea, and glycerol were purchased from Aladdin Reagent Co., Ltd. (Shanghai, China). MCC (20–100 μm, molecular weight 162.06$_n$, $n \approx 220$) was purchased from Guangfu Chemical Reagent Co., Ltd. (Tianjin, China). Xylan (purity: $\geq$95%) was purchased from Solarbio Science & Technology Co. Ltd. (Beijing, China). Alkaline lignin was purchased from Yuanye Bio-Technology Co., Ltd. (Shanghai, China). Copper(II) ethylenediamine complex (1 M, molecular weight 121.63) for determining the degree of polymerization of cellulose was purchased from Guangfu Chemical Reagent Co., Ltd. Sulfuric acid, sodium hydroxide, acetic acid, and acetone were purchased from Kermel Chemical Reagent Co., Ltd. (Tianjin, China). All chemicals were of analytical purity.

Preparation of DES: ChCl was coupled with urea, oxalic acid dihydrate, and glycerol to obtain three kinds of DESs. The binary mixtures with a molar ratio of 1:1 were allowed to react under reduced pressure at 80 °C for 1 h. Kamlet–Taft solvatochromic parameters were adopted to characterize these DESs (Note S2 in the Supporting Information).

Fractionation of WL with DES: WL (0.5 g) was mixed with DES (10 g), and the mixture was heated in an oil bath or with a microwave. The processing conditions for the oil-bath treatment were 80 and 110 °C for 9 h. Microwave-assisted heating was conducted with an XH-800C Microwave Accelerated Reaction System (Xianghu Technology Co., Ltd., Beijing, China). The temperature was raised from room temperature to 80 °C over 2 min and held for 1, 3, or 8 min. After treatment, the resultant dark-brown liquid was poured

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into acetone/water (200 mL, 1:1 *v/v*) anti-solvent, and the mixture was stirred for 1 h. The mixture was vacuum-filtered and washed with the same solvent three times. After filtration, the residue was denoted CR. The liquid fractions were further heated in a rotary evaporator at 60 °C to remove acetone, and the concentrated solutions were also filtrated to retain solids, which were denoted LF. The leached CR and LF were denoted $DES_{OB80\text{-}9}$, $DES_{OB110\text{-}9}$, $DES_{Mw80\text{-}3}$, $DES_{Mw80\text{-}5}$, and $DES_{Mw80\text{-}10}$.

Characterization techniques: 2D HSQC NMR spectra were acquired with an Agilent Pro Pulse 500 MHz spectrometer (Agilent Technologies, Santa Clara, CA, USA), and recorded in the gradient HSQC adiabatic version. The 90 mg samples were dissolved in [D_6]DMSO (0.5 mL). The spectral widths were 20000 and 5000 Hz for the ^{13}C and ^{1}H dimensions, respectively. The number of collected complex points was 1024 for the ^{1}H dimension with a recycle delay of 1.5 s, the number of transients was 128, and 256 time increments were recorded in the ^{13}C dimension. Prior to Fourier transformation, the data matrices were zero-filled to 1024 points in the ^{13}C dimension.

Molecular weights were measured by a Waters 2695 GPC (Milford, MA, USA) with a differential refractive index detector (Waters 2414). The column was operated at ambient temperature and eluted at a flow rate of 1 mL min^{-1}. M_n and M_w distributions were calibrated by using polystyrene standards.

Pyrolysis of 100 μg MWL and $DES_{Mw80\text{-}3}$ LF was performed with a CDS5200 pyrolyzer connected to an Agilent 6890N/5973i GC/MS system. The pyrolysis was performed at 500 °C for 30 s (20 °C ms^{-1}). The oven temperature was programmed from 40 to 230 °C at 5 °C min^{-1}. Helium (99.999%) was used as carrier gas (1 mL min^{-1}). The compounds were identified by comparing their mass spectra with those of the Wiley and NIST libraries. Peak molar areas were calculated for the LF, the summed areas were normalized to 100, and the data for two repetitive analyses were averaged and expressed as percentages.

FTIR spectra were recorded with a Nicolet Magna 560 instrument (Thermo Fisher, Waltham, MA, USA) with a diamond attenuated total reflectance (ATR) attachment. The spectra were recorded in ATR mode, and the data were recorded over the range from 650 to 4000 cm^{-1} in absorbance mode with 32 scans per spectrum and a resolution of 4 cm^{-1}. The background was subtracted before the analysis.

XRD patterns were obtained with an X-ray diffractometer (D/max 2200, Rigaku, Tokyo, Japan) with Ni-filtered CuKα radiation ($\lambda = 0.154$ nm) at 40 kV and 30 mA. The diffraction data were collected in the range $2\theta = 5$–40° at a scanning rate of 4° min^{-1}. The relative crystallinity index (CrI) was estimated by the Segal method by using the following equation: $CrI = (I_{002} - I_{am})/I_{002} \times 100\%$, in which I_{002} is the peak intensity of the (0 0 2) lattice diffraction at $2\theta \approx 22.6^\circ$, and I_{am} is the diffraction intensity of amorphous fraction at $2\theta \approx 18^\circ$.

TG analysis of the samples was performed with a Pyris 6 (PerkinElmer, Waltham, MA, USA) at a heating rate of 10 °C min^{-1} under nitrogen atmosphere. The equipment was calibrated through baseline correction before testing.

TEM observations on the samples were performed with a JEM-2100 microscope at an accelerating voltage of 200 kV. A droplet (5 μL) of diluted slurry was dropped on the carbon-coated electron microscopy grid, and then was negatively stained with 1 wt% phosphotungstic acid solution to enhance the image contrast. The size of the nanocellulose was measured from the TEM images by using a TDY-V5.2 image analysis system (Tianhong Precision Instrument Co. Ltd., Beijing, China).

The saccharide analysis was conducted with a HPAEC system (Dionex ICS3000, US) with a pulsed amperometric detector, an AS50 autosampler, a CarbopacTM PA-20 column (4×250 mm, Dionex), and a PA-20 guard column (3×30 mm, Dionex). The saccharides were first precipitated by adding a threefold volume of ethanol and then analyzed by HPAEC. The composition of saccharides was quantified by using standard solutions of neutral carbohydrates (glucose, xylose, rhamnose, mannose, arabinose, and galactose).

Yield of HMF was determined by HPLC with acetonitrile/water (10:90) as mobile phase (0.8 mL min^{-1}). The measurement wavelength was 285 nm.

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Keywords: biomass · deep eutectic solvents · lignin · lignin–carbohydrate complexes · microwave chemistry

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